

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name			
		Mr. S Naresh Kumar	
		Ms. B. Swathi	
		Dr. Sasanko Shekhar Gantayat	
		Mr. Md Sallauddin	
		Dr. Mathivanan	
		Mr. Y Srikanth	
		Ms. N Shilpa	
		Dr. Rishabh Mittal (Coordinator)	
		Dr. R. Prashant Kumar	
		Mr. Ankushavali MD	
		Mr. B Viswanath	
		Ms. Sujitha Reddy	
		Ms. A. Anitha	
		Ms. M.Madhuri	
		Ms. Katherashala Swetha	
Ms. Velpula sumalatha			
Mr. Bingi Raju			
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week5 – Monday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 9.1(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab Experiment: Documentation Generation -Automatic documentation and code comments	Week5 - Monday	

	Lab Objectives 1. To understand automatic documentation generation. 2. To generate code comments and docstrings using AI tools. 3. To learn the importance of documentation in software development.	
	Lab Outcomes 1. Students will be able to generate documentation automatically for code. 2. Students will be able to add clear comments and docstrings to programs. 3. Students will be able to improve code readability and maintainability using documentation.	
	Problem 1: Consider the following Python function: <pre>def find_max(numbers): return max(numbers)</pre> Task: • Write documentation for the function in all three formats: (a) Docstring (b) Inline comments (c) Google-style documentation • Critically compare the three approaches. Discuss the advantages, disadvantages, and suitable use cases of each style. • Recommend which documentation style is most effective for a mathematical utilities library and justify your answer.	

	Problem 2: Consider the following Python function: <pre>def login(user, password, credentials): return credentials.get(user) == password</pre> Task: 1. Write documentation in all three formats. 2. Critically compare the approaches. 3. Recommend which style would be most helpful for new developers onboarding a project, and justify your choice.	
	Problem 3: Calculator (Automatic Documentation Generation) Task: Design a Python module named calculator.py and demonstrate automatic documentation generation. Instructions: 1. Create a Python module calculator.py that includes the following functions, each written with appropriate docstrings: - o add(a, b) – returns the sum of two numbers - o subtract(a, b) – returns the difference of two numbers - o multiply(a, b) – returns the product of two numbers - o divide(a, b) – returns the quotient of two numbers 2. Display the module documentation in the terminal using Python’s documentation tools. 3. Generate and export the module documentation in HTML format using the pydoc utility, and open the generated HTML file in a web browser to verify the output.	

	Problem 4: Conversion Utilities Module Task: 1. Write a module named <code>conversion.py</code> with functions: ○ <code>decimal_to_binary(n)</code> ○ <code>binary_to_decimal(b)</code> ○ <code>decimal_to_hexadecimal(n)</code> 2. Use Copilot for auto-generating docstrings. 3. Generate documentation in the terminal. 4. Export the documentation in HTML format and open it in a browser.	
	Problem 5 – Course Management Module Task: 1. Create a module <code>course.py</code> with functions: ○ <code>add_course(course_id, name, credits)</code> ○ <code>remove_course(course_id)</code> ○ <code>get_course(course_id)</code> 2. Add docstrings with Copilot. 3. Generate documentation in the terminal. 4. Export the documentation in HTML format and open it in a browser.	